

SuperKart Sales Forecasting

Problem Understanding and Proposed Approach :-

1. Problem Statement

SuperKart is a supermarket chain that wants to predict the future sales revenue of different outlets for the upcoming quarter. The objective is to build a machine learning model that can estimate the total sales generated by products across different stores based on historical sales data.

This prediction will help the organization in:

- Inventory planning
- Supply chain optimization
- Demand forecasting
- Region-wise sales strategy
- Resource allocation

The target variable in this problem is - **Product_Store_Sales_Total**

2. Understanding the Dataset

The dataset consists of product-level and store-level features.

Product-related features:

- Product_Id
- Product_Weight
- Product_Sugar_Content
- Product_Allocated_Area
- Product_Type
- Product_MRP

Store-related features:

- Store_Id
- Store_Establishment_Year
- Store_Size
- Store_Location_City_Type

- Store_Type

Target variable:

- Product_Store_Sales_Total

Which tells us this is a **supervised regression problem** because the output variable is continuous.

3. Problem Challenges

Before building the model, some challenges in the data need to be addressed:

A. Mixed Data Type - The dataset contains both numerical and categorical variables.

Example:

- Numerical: Product_Weight, Product_MRP
- Categorical: Store_Type, Product_Type

Therefore preprocessing strategies are required.

B. Missing Values - Some fields may contain missing values, especially product weight.

C. Feature Relationship - Sales are influenced by multiple interacting factors.

Example:

- Product MRP + Store Size
- Product Type + City Tier
- Store Type + Product Category

These nonlinear relationships must be captured.

D. High Cardinality ID - Product_Id and Store_Id may contain useful hidden patterns but cannot be directly used without proper handling.

4. Proposed Approach

The complete approach will be divided into 6 stages.

Stage 1: Data Exploration (EDA)

The first step will be understanding the dataset.

Tasks:

- Check shape of dataset
- Check data types
- Check missing values
- Analyze feature distributions
- Detect outliers
- Understand correlations

Stage 2: Data Preprocessing

A. Missing Value Treatment

- Numerical columns → median imputation
- Categorical columns → mode imputation

B. Feature Encoding

For baseline models:

- Label Encoding / One-Hot Encoding

For CatBoost:

- Native categorical handling

C. Feature Engineering

Planned engineered features:

Store Age

$\text{Store_Age} = \text{Current Year} - \text{Store_Establishment_Year}$

Reason:

Older stores may have established customer bases.

Product Category Prefix

Extract category prefix from Product_Id.

Reason:

The first two letters may represent product groups.

MRP Buckets

Convert Product_MRP into price ranges.

Reason:

Helps mo

del price segmentation.

Sales Density Features

Combine allocated area and store size.

Reason:

May indicate display efficiency.

Stage 3: Baseline Model Building

Initial models will be built for benchmarking.

Model 1: Linear Regression

Purpose:

- Understand linear relationships
- Create baseline RMSE

Limitation:

Cannot capture nonlinear patterns.

Model 2: Random Forest Regressor

Purpose:

- Capture nonlinear relationships
- Learn feature interactions

Improvement over Linear Regression:
Handles complexity better.

Limitation:
Trees are independent and may not optimally correct errors.

Stage 4: Advanced Boosting Models

Model 3: XGBoost

Purpose:

- Sequentially correct errors
- Improve prediction accuracy
- Learn complex interactions

Advantages:

- Strong for numerical patterns
- Regularization support

Limitation:
Needs encoded categorical variables.

Model 4: CatBoost

Purpose:

- Handle categorical variables efficiently
- Reduce preprocessing
- Improve generalization

Advantages:

- Native categorical handling
- Ordered boosting reduces leakage

Reason for choosing:
The dataset has many categorical variables, making CatBoost highly suitable.

Stage 5: Ensemble Final Model

To improve performance further, stacking will be used.

Base models:

- XGBoost
- CatBoost

Meta-model:

- Linear Regression

Workflow:

1. Train XGBoost
2. Train CatBoost
3. Generate out-of-fold predictions
4. Train Linear Regression on these predictions
5. Produce final prediction

Reason:

This combines:

- XGBoost's strength in numerical interactions
- CatBoost's strength in categorical handling

6. Evaluation Metrics

The models will be evaluated using:

RMSE (Root Mean Squared Error)

Measures average prediction error.

Lower is better.

MAE (Mean Absolute Error)

Measures average absolute difference.

Lower is better.

R² Score

Measures how much variance is explained.

Higher is better.

7. Final Expected Outcome

The final objective is to build a robust regression model capable of accurately forecasting product-store sales.

Expected deliverables:

- Data analysis insights
- Cleaned dataset
- Multiple model comparisons
- Final ensemble model
- Performance evaluation
- Business recommendations based on feature importance

8. Conclusion

This project will follow a structured machine learning pipeline starting from simple models and gradually moving toward advanced ensemble techniques. The final stacked model is expected to provide the most accurate sales predictions by combining the strengths of multiple boosting models.